

DEMO DATA â Drilling Anomaly Report (Torque Spike & Drag)

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DEMO DATA - OIL INDIA LIMITED | DRILLING OPERATIONS GROUP
DRILLING ANOMALY REPORT - HIGH TORQUE & DRAG INVESTIGATION

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Well: Duliajan-NWIS-01 | Well ID: OIL-NWIS-ACT-001 | Field: Duliajan
Depth: 2780m | Formation: Barail Group (Interbedded Sand/Shale)
Inclination: 28.4 degrees | Azimuth: 142 degrees

DRILLING EVENT: TORQUE SPIKE at 2780m (Severity: MEDIUM)
Description: Sudden torque spike from 14 kN.m to 28.5 kN.m in under 3 minutes with erratic rotary stalling.
Cause: Cuttings bed accumulation in 28-degree build section and localized micro-dogleg.
Mitigation: Backed off WOB from 16t to 8t. Circulated high-viscosity pill sweep and performed back-reaming pass.
Outcome: Hole cleaned effectively. Rotary torque stabilized at 15 kN.m; resumed drilling safely after 1.5 hrs NPT.
NPT Hours: 1.5

DRILLING PARAMETERS AT EVENT:
WOB: 16 t | RPM: 45 | Torque: 28.5 kN.m | ROP: 2.5 m/hr
MW: 1.36 g/cc | SPP: 255 bar | Flow Rate: 2100 L/min

LESSONS LEARNED: Circulate high-viscosity hole cleaning sweep every stand when inclination exceeds 25 degrees.
BEST PRACTICE: Limit rotary speed to 50 RPM and maintain reaming speed <10 m/hr in deviated Barail section.

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